

The Impact of Graduated Backpack Loading on Sagittal Plane Kinematics and Joint Kinetics during Gait

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Abstract

This study investigated how progressively increasing backpack loads affect sagittal-plane gait mechanics in healthy young adults during overground walking. Fifteen participants completed walking trials under three loading conditions (0%, 10%, and 20% body weight (BW)) while trunk lean, hip angle, and knee angle were analyzed using two-dimensional motion capture in Kinovea. The primary goal was to determine whether increasing external load produced measurable alterations in gait kinematics and whether a specific loading threshold could be identified at which movement patterns deviated significantly from baseline walking. Results demonstrated consistent numerical trends across all measured variables as load increased. Mean trunk forward lean increased from $7.96 \pm 4.22^\circ$ at 0% BW to $9.24 \pm 5.03^\circ$ at 10% BW and $12.07 \pm 5.12^\circ$ at 20% BW, indicating progressively greater forward inclination under heavier loads. Similarly, minimum knee angle increased from 123.79° to 129.71° , suggesting a more extended knee posture during loaded walking, while minimum hip angle decreased from 145.50° to 137.31° , reflecting increased hip flexion as backpack load increased. Although these findings demonstrated clear directional changes consistent with compensatory strategies used to maintain balance and center of mass during load carriage, repeated-measures ANOVA revealed no statistically significant differences between loading conditions for trunk, hip, or knee kinematics. Considerable inter-subject variability was observed, with some participants demonstrating pronounced postural adaptations while others maintained movement patterns similar to unloaded walking. These findings generally align with previous biomechanical literature reporting increased trunk lean and altered lower-extremity mechanics during load carriage, and supports substantial individual variability. Strengths of this study included the repeated-measures design, ecologically valid overground walking protocol, and simultaneous examination of multiple sagittal-plane variables. Limitations included the small sample size, use of 2D motion analysis, and self-selected walking speeds, all of which likely reduced statistical power and measurement precision. Clinically, these results suggest that adapted backpack loads up to 20% BW produce observable but highly individualized gait adaptations in healthy young adults rather than a universal threshold.

Introduction

Clinical Problem

Gait with variable loading presents a significant clinical challenge, as the addition of external loads to the body may alter posture and movement mechanics, increasing trunk forward lean and imposing greater strain on the joints of the lower extremities. These loads are exemplified across a wide range of everyday contexts. For example, a student carrying a school backpack may effectively add approximately 4 kg or more to their body mass [1]. In comparison, an occupational worker such as a restaurant busser may carry upwards of 5 kg of plates and

dishware across an extended shift. Regardless of context, such loads act on the lower extremities and the musculature proximal to the shoulder joint, with their biomechanical consequences shaped by how the load is distributed across the trunk. When experienced repeatedly over time, these mechanical demands can drive structural and functional adaptations within the musculoskeletal system, altering habitual movement patterns and postural alignment in ways that may persist beyond the loading event itself. This is of particular clinical concern for populations who carry loads regularly, including schoolchildren, military personnel, and manual laborers. Among these populations, cumulative loading has been associated with elevated injury risk and lasting musculoskeletal consequences [2]. A thorough kinematic analysis of load-induced gait changes is therefore essential for establishing safe load thresholds, developing strategies for treatment, and mitigating the long-term risks of postural deviation and musculoskeletal strain.

Previous Work

A substantial body of research has examined the biomechanical and physiological effects of backpack load carriage, consistently demonstrating that external loads influence gait, posture, and overall movement mechanics. However, the clinical problem remains incompletely understood due to inconsistent findings and methodological limitations across studies. A narrative review by Genitrini et al. highlights that backpack load significantly alters gait kinematics, posture, and metabolic cost, with outcomes strongly dependent on factors such as load weight, pack design, and fit [3]. Importantly, the review emphasizes that these interacting variables make it difficult to define a single “optimal” loading condition, underscoring that the problem is not yet resolved. Similarly, dynamic biomechanical analyses have shown that load carriage increases joint loading and forces on the trunk and lower extremities during walking; this study leaves concerns about long-term injury risk unaddressed [4,5]. Other studies demonstrated that increasing backpack weight reduces joint range of motion and alters gait patterns, while adjustments such as hip belt tension can redistribute load [6]. However, the effects of these modifications on gait mechanics remain inconsistent, indicating an incomplete understanding of how load distribution strategies influence movement. Research in children further supports the association between heavier loads and postural deviations, particularly when loads exceed 10–15% of body weight [7]. Further studies reported that loads exceeding 15–20% of body weight significantly increase trunk forward lean, temporal-spatial gait parameters, joint moments, and respiratory rate in children, indicating elevated physical stress, while loads below 10% have minimal effects [7, 8, 9]. It should be noted that several studies were conducted under controlled or simulated conditions, including simulations [4] and treadmill-based experiments [5, 6, 8]. While these allow precise control of variables, they reduce ecological validity, as walking velocity and acceleration do not reflect real-world conditions. Notably, considerable variability between test populations has been observed, and authors have called for further investigation into the cumulative musculoskeletal effects of load carriage to better inform safe load recommendations. Collectively, these studies demonstrate that while backpack load carriage clearly impacts gait mechanics, posture, and physiological responses, findings are often

inconsistent and limited by narrow experimental designs. As a result, the overall impact of different load configurations on walking mechanics remains unclear, justifying the need for further research using more comprehensive and ecologically valid approaches.

What Needs To Be Done

Despite extensive research on backpack load carriage, questions remain about how heavier loads alter gait mechanics in healthy young adults. Prior studies have focused on only one factor at a time, such as how much weight is carried or how the weight is distributed, even though real backpack use involves several factors occurring simultaneously [3]. Existing literature also focuses on children or military personnel, leaving a gap in understanding how young adults, who frequently carry heavy backpacks in daily life, adapt their gait under increasing load [8]. Even studies that tested loads between 10-20% of body weight have reported mixed results about how backpack weight affects joint angles and trunk posture, partly because many focused more on forces than on movement patterns and used different walking speeds or testing setups [2, 10, 11]. These inconsistencies make it unclear whether heavier backpacks cause predictable changes, like leaning forward more or changing hip movement, or whether people respond differently from one another depending on the weight. This creates a need for a controlled study that focuses specifically on kinematic changes in healthy young adults as load increases. By examining how different backpack loads affect lower-limb joint angles and trunk posture, this study aims to identify which movement variables are most sensitive to load and determine whether heavier backpacks reliably alter gait mechanics.

Study Goal

The overarching goal of this study is to quantify how the addition of external load, expressed as a percentage of an individual's body mass, influences the biomechanics of both the trunk and lower extremities during level walking. To address the gaps identified in the existing literature, three specific aims were established. The first aim is to determine a threshold of additional body mass percentage at which musculoskeletal joint loading increases to a degree that meaningfully elevates injury risk, with the expectation that loads beyond a certain percentage of body weight will produce disproportionate increases in joint stress relative to lighter loads. The second aim is to evaluate changes in sagittal plane kinematics across loading conditions, specifically examining how progressive load increases affect trunk forward lean, hip flexion angle, knee angle, and step length, which are variables prior work has shown to be sensitive to load but inconsistently characterized across studies. The third aim is to determine how added load influences joint moments and ground reaction forces at the hip, knee, and ankle, with the hypothesis that increasing body-borne load will produce measurable and load-dependent increases in these kinetic variables that scale with the magnitude of load applied. Together, these aims are designed to produce a more complete and internally consistent characterization of

load-induced gait changes in healthy young adults, with direct implications for defining safe load thresholds in everyday and occupational settings.

Methods

Experimental Design

This study utilizes a repeated-measures experimental design to assess the effects of progressively increasing backpack loads on gait kinematics during walking. Each participant served as their own control and completed trials under four loading conditions: 0%, 10%, and 20% of their body weight. This within-subject design allows for direct comparison of biomechanical changes across loading conditions while minimizing inter-individual variability.

Prior to data collection, baseline demographic and anthropometric data were recorded, including age, sex, body weight, height, leg length, and trunk length. Participants were then fitted with a standard backpack, and their initial weight was recorded with the empty backpack to establish the baseline (0% body weight condition). Additional loads are then applied using standardized dumbbell weights (2.5, 5, 10, and 25 lbs), which are placed inside the backpack and adjusted to achieve the target percentages of body weight for each condition.

Data collection took place in a controlled environment using a single camera mounted on a tripod to capture two-dimensional motion in the sagittal plane. A defined walking path was marked using tape on the floor, including designated zones for acceleration, steady-state walking, and deceleration. This approach ensures that gait data are collected during steady-state walking while still approximating overground, real-world conditions.

Colored markers were placed on anatomical landmarks on one side of the body, including the shoulder, hip, lateral knee, and lateral ankle. To reduce variability due to arm swing and improve tracking of upper body motion, participants were instructed to keep their hands stationary by holding the backpack straps throughout each trial. Participants holding the straps also improved the visibility of the hip marker during data collection.

Each participant completed three walking trials per load condition, resulting in a total of nine trials per participant. Trials were conducted in a fixed order of increasing load to promote participant safety and gradual adaptation to added weight. Adequate time was provided between trials as needed to prevent fatigue.

All trials were recorded using a single camera, and video data were saved for later analysis. Kinematic data were extracted from recorded trials using Kinovea, with particular focus on steady-state gait cycles within the capture region. Participant data, including trial conditions and measurements, were recorded.

To ensure data quality, potential sources of error were addressed during collection. If markers became obstructed or detached, trials were repeated after correcting placement. This design enables consistent and reliable data collection while maintaining relevance to real-world walking conditions.

Subject Selection

This study intended to select able-bodied individuals aged between 18 and 23 without a recent history of musculoskeletal injury that could influence gait or load-carrying mechanics. Inclusion criteria were designed to ensure a healthy, physically capable population representative of average young adults. Exclusionary criteria included an inability to walk independently, difficulty or discomfort when lifting approximately 20% of body weight, or unwillingness to participate. Participants were excluded if they experienced major pain during testing or if they rescinded consent at any time during the experimental testing procedure or subsequent data analysis. This ensured both participant safety and ethical compliance throughout the study. Subjects will be excluded from data analysis due to poor marker tracking, video quality, or subjects not holding backpack straps appropriately.

Recruitment targeted $N \geq 15$ participants by informally reaching out to friends, classmates, roommates, and other peers in Spencer Lab. This approach allowed for access to a population that is both accessible to the researchers and likely to meet the study's inclusion criteria. Although a smaller convenience sample was used, it is appropriate for an initial biomechanical analysis focused on identifying general trends in load carriage effects.

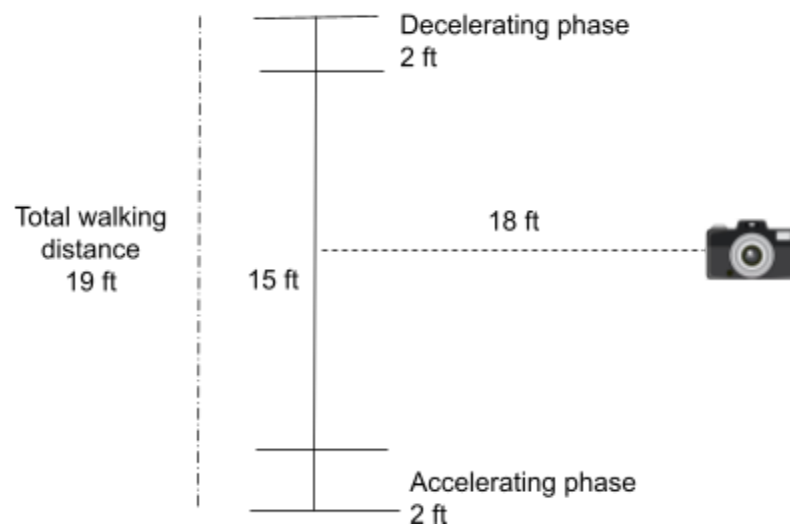
Before testing, all prospective participants underwent an informed consent process. Each participant was provided a written consent document outlining the study's purpose, procedures, potential risks, and their rights as a participant, including the ability to withdraw at any time. The personal data recorded for this study about the participant was kept private and undisclosed. Participants were allowed to ask questions and would only proceed once written consent had been obtained. This process ensured adherence to standard human subjects research protocols and reinforced the ethical integrity of the study. Participant demographics were recorded at the start of the testing session and included age, sex, height, leg length, body weight, and self-paced walking speed. The group statistics are presented in Table 1.

Table 1. Group Participant Demographics

Variable	Mean	Range
Age (years)	21 years	20 - 22 years
Sex (M/F)	M = 5, F = 10	
Height (cm)	168.5 cm	152.4 - 183 cm
Leg Length (cm)	90 cm	85 - 100 cm
Body Weight (lbs)	151 lbs	104 - 198 lbs
Walking Speed (m/s)	4.3 ft/s	3.54 - 5.34 ft/s

Test Setup

This experiment is designed to be straightforward and as close to real-world walking conditions as possible. This study will be conducted outside on a flat brick path to avoid walking with laboratory constraints. Individuals will walk a total of 19 feet at their own walking pace. There will be a 2-foot acceleration and deceleration phase, and the total recorded walking distance will be 15 feet, as shown in **Figure 1**. These distances will be marked with tape to ensure all patients walk the same path. This setup allows participants sufficient space to reach a natural gait and complete approximately three full gait cycles. A single camera mounted on a tripod was placed 18 feet away from the walking path and positioned perpendicular to the walking path to capture motion in the sagittal plane. The camera frame was aligned to include only the 15-foot data collection zone, ensuring that only steady-state walking was analyzed (**Figure 2**).

**Figure 1:** Mock-up sketch of the camera and walking setup being used in this study

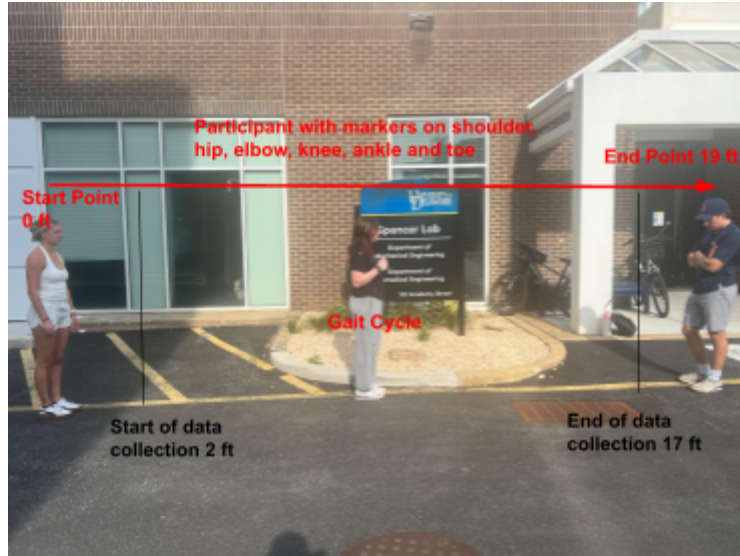


Figure 2: Real setup to demonstrate the distance and frame of the camera.

Colored markers will be placed on anatomical landmarks on the participants' dominant side, including the shoulder, hip, lateral knee, and ankle. These markers will be used to track the joint angles during the gait on Kinovea. Specifically, trunk lean angle was determined using the shoulder, hip, and knee markers; knee joint angle was calculated using the hip, knee, and ankle markers; and ankle angle was determined using the knee, ankle, and foot markers.

Participants will wear a Jansport brand school-style backpack, shown in **Figure 3**, throughout all trials. The backpack will be adjusted to fit each participant as intended. Initial measurements included body weight (with backpack), height, leg length, and sex. Additional weight was added to the backpack using standardized weights (2.5, 5, 10, and 25 lbs) to reach loading conditions of 0%, 10%, and 20% of each participant's body weight. Each participant completed a total of nine walking trials: three trials at each loading condition (0%, 10%, and 20% body weight). Trials were conducted in sequential order, starting with no added weight and progressing to the heaviest load. Participants were instructed to walk at a self-selected, comfortable pace.



Figure 3: Jansport style backpack that will be used in all trials. Shoulder straps will be adjusted to fit each participant properly. Weight is negligible to the study.

Numerical data will be recorded and saved in a Google Sheet, and visual data will be recorded and saved as .mp3 video files. The visual data is organized by subject, trial, and body weight %. These videos are uploaded into Kinovea for further analysis, and all kinematic and kinetic data from the Google sheet and Kinovea will be compared and statistically analyzed.

Outcome Measures

Kinematic data will be collected from video recordings captured with an iPhone 15 camera at 60 frames per second, with sufficient resolution to clearly identify gait events and joint angle changes. All recordings will be imported into Kinovea for digitizing and angle measurement. For each participant, the best recording of the three recorded for each loading condition, considering marker quality and analysis, will be used for final data analysis. For each load condition (0%, 10%, and 20% BW), at least three consecutive gait cycles will be analyzed once the participant reaches steady-state walking. To avoid the influence of starting and stopping, only the middle cycles of each recording will be used. Heel strike and toe-off will be identified visually in Kinovea to define the gait cycles. Joint angles will be calculated from segment lines created in software: hip angle as the angle between the trunk and thigh segments in the sagittal plane, knee angle as the angle between the thigh and shank, and trunk angle as the forward-lean angle of the trunk segment. The primary variables that will be recorded include peak trunk forward lean during stance, as well as peak hip flexion/extension, and knee flexion during the gait cycle.

Data Analysis

A repeated measures design will be used, with each participant completing all three backpack load conditions. The independent variable will be the load condition (0%, 10%, & 20% BW), and the dependent variables will be the kinematic outcomes described in “Outcome Measures”. For each participant and each load condition, peak values for the trunk forward lean, hip flexion/extension, and knee flexion will be identified within the selected gait cycles. These peak values will be averaged across the cycles to create representative data for each condition.

All processed data will be entered into JMP in a long-format table, with each row representing one participant under one load condition. A one-way repeated measures ANOVA will be performed separately for each kinematic variable to determine whether backpack load produces significant changes in movement patterns. If the ANOVA shows a significant main effect of load, post-hoc pairwise comparisons will be used to identify which load conditions differ from one another. Statistical significance will be set at $\alpha = 0.05$ for all tests. Along with p-values, effect sizes will be reported to describe the magnitude of differences for ANOVA results. Any trial that cannot be analyzed due to poor visibility or incomplete gait cycles will be excluded, and the remaining valid cycles will be used to compute the participant's average for that condition.

Results

To assess whether the additional load influenced the kinematic indicators commonly associated with increased mechanical demand, the combined behavior of trunk, hip, and knee angles was examined across conditions. Participants displayed a consistent pattern in which trunk lean increased, hip angle decreased, and knee angle increased as the load increased from 0% to 20% BW. One participant's posture is shown in **Figure 4**, demonstrating the visual forward trunk lean. This pattern showed a change in overall posture with greater forward inclination of the trunk, more extended knees, and a more flexed hip position during the stance phase. However, the magnitude of these changes varied considerably across individuals, with some participants showing more pronounced adjustments to the load and others maintaining normal movement patterns similar to their unloaded gait. Statistical analysis confirmed that none of these variables demonstrated a significant effect of load, and no pairwise comparisons reached significance. The graphs in **Figure 5** summarize the raw data and show each angle (trunk, hip, and knee) over the course of the entire walking path. They display the maximum and minimum angles, representing typical gait patterns. . It should be noted that Kinovea measures complementary hip and knee angles and may switch between positive and negative values. Therefore, to normalize the data, the graphs represent the difference between 180° and the absolute value of the Kinovea measurement.

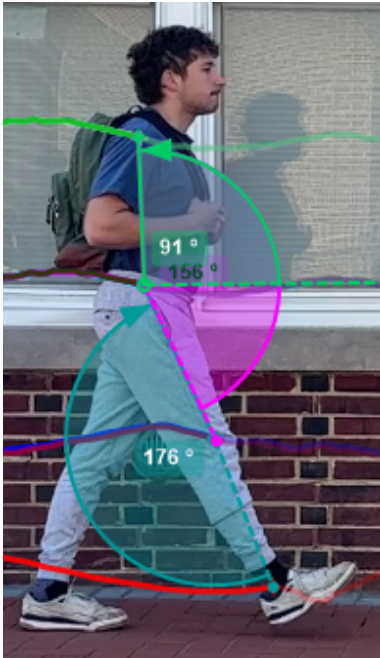


Figure 4A



Figure 4B



Figure 4C

Figure 4: One subject example of a visible trunk lean as backpack load increases. It's important to note that not all subjects demonstrated the same increasing lean, which is what leads to statistical insignificance.

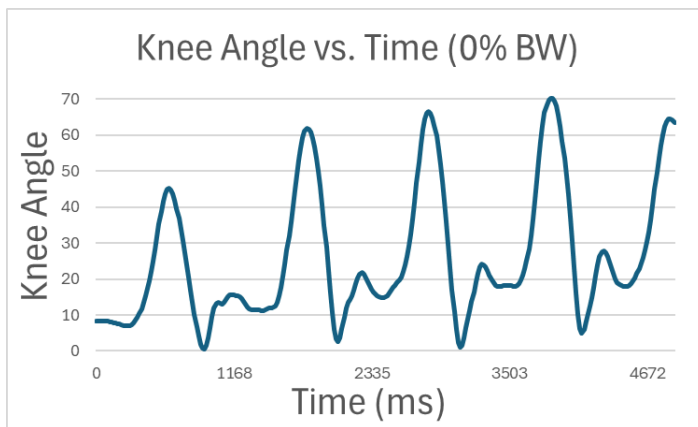


Figure 5A

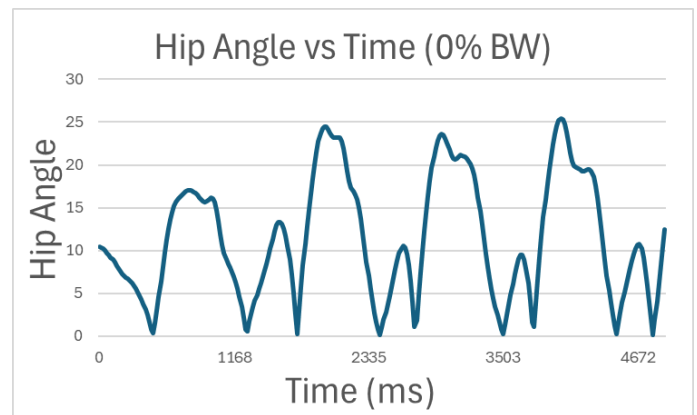


Figure 5B

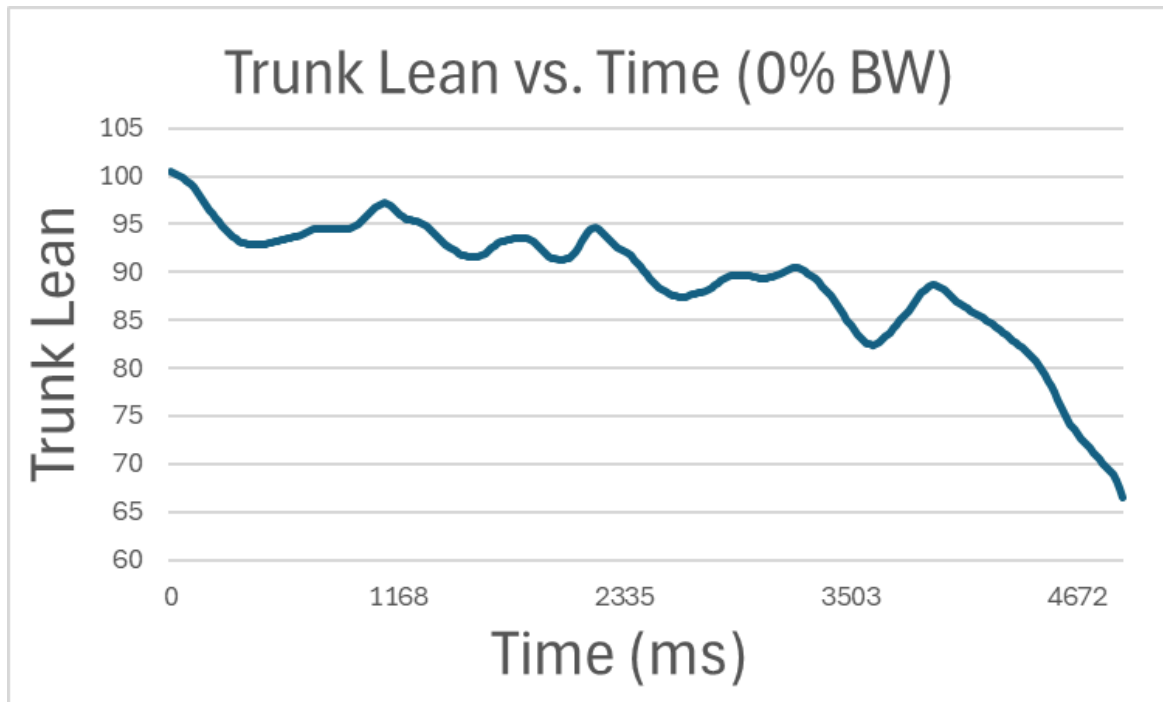


Figure 5C

Figure 5: Graphs generated by Kinovea give a clear visualization of gait patterns during the walking path. Figure 5A and 5B show knee angle and hip angle fluctuating depending on where the subject was in their gait cycle. Figure 5C shows that the trunk angle decreases with time, proving strenuous loads lead to poor walking mechanics.

To evaluate whether a specific load percentage produced a measurable deviation from baseline gait, trunk forward lean was examined across the three loading conditions. Participants exhibited a relatively upright posture during unloading walking, with trunk lean values clustered around the lower end of the measured range. When the load increased to 10% BW, several participants demonstrated moderate increases in forward inclination, while others showed minimal change, resulting in a wider distribution of values. At 20% BW, the group displayed the largest spread in trunk lean, with some individuals showing substantial increases and others remaining close to their baseline posture. Mean trunk forward lean was $7.96 \pm 4.22^\circ$ at 0% BW, $9.24 \pm 5.03^\circ$ at 10% BW, and $12.07 \pm 5.12^\circ$ at 20% BW, indicating a numerical increase across conditions. Despite these shifts, repeated measures ANOVA indicated no statistically significant difference between the three conditions (F ratio = 2.30, $p = 0.12$), and no pairwise comparison reached significance. Thus, although individual participants exhibited noticeable changes in posture at higher loads, statistical testing did not identify a specific load percentage at which trunk posture diverged from the unloaded condition. These distributions are shown in **Figure 6**.

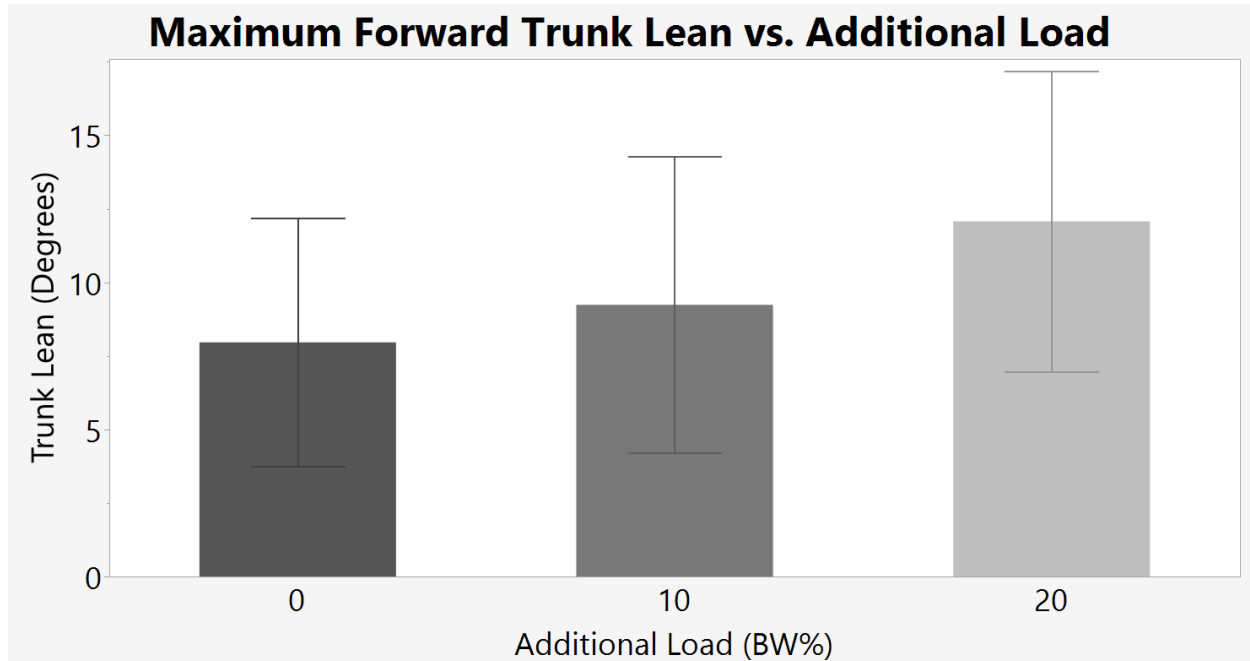


Figure 6: Mean maximum forward trunk lean increased with greater additional backpack load (0%, 10%, and 20% BW), although the effect was not statistically significant.

To characterize how lower-extremity kinematics behaved across progressively increasing loads, knee and hip angles were examined alongside trunk posture. Minimum normalized knee angle demonstrated a gradual downward shift across conditions, with the group mean decreasing from 56.21° at 0% BW to 53.47° at 10% BW and 50.29° at 20% BW. A graph demonstrating this data can be seen in **Figure 7**. This pattern indicates a tendency toward a more extended knee position during loaded walking, although substantial variability was present across participants. Minimum normalized hip angle showed the opposite trend, increasing from 34.5° at 0% BW to 36.17° at 10% BW and 42.69° at 20% BW, indicating a greater degree of hip flexion under heavier loads. These values are shown in **Figure 8**. It's important to note that Kinovea measures the complementary angles of the knee and hip; therefore, the data in **Figure 7 and 8** represents the difference between 180° and the Kinovea value provided. Despite these changes, repeated measures ANOVA revealed no significant main effect of load on the knee, hip, or trunk angle ($p = 0.73$, $p = 0.78$, $p = 0.12$, respectively).

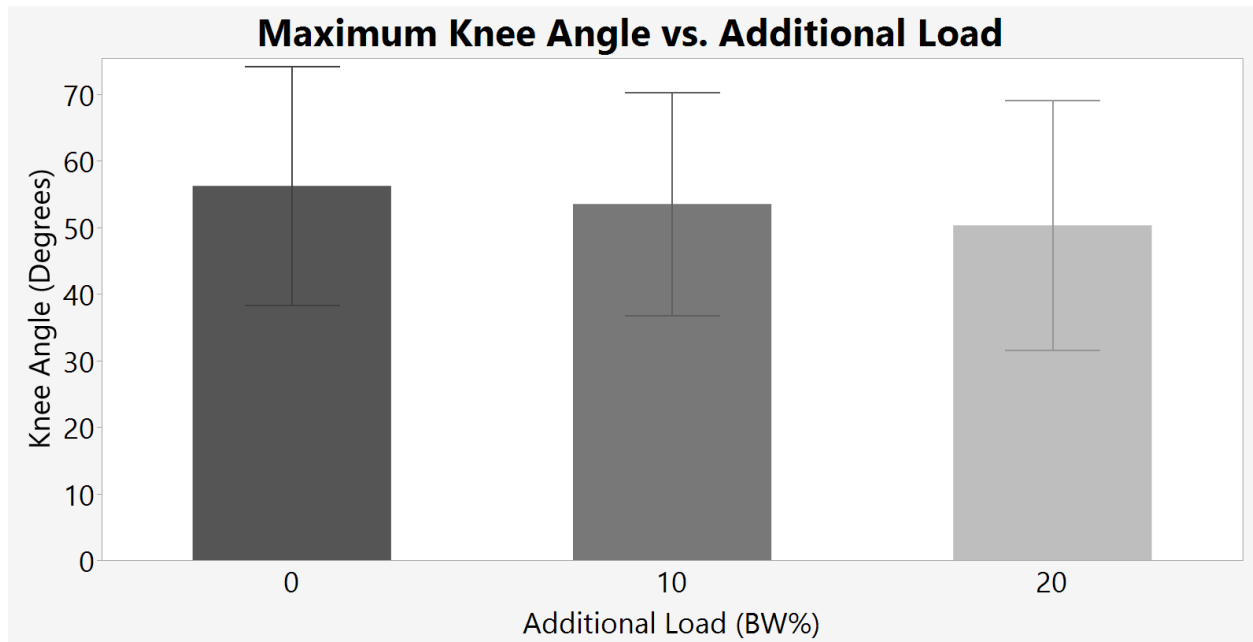


Figure 7: Mean minimum knee angle decreased slightly with backpack load. Data did not show statistically significant results.

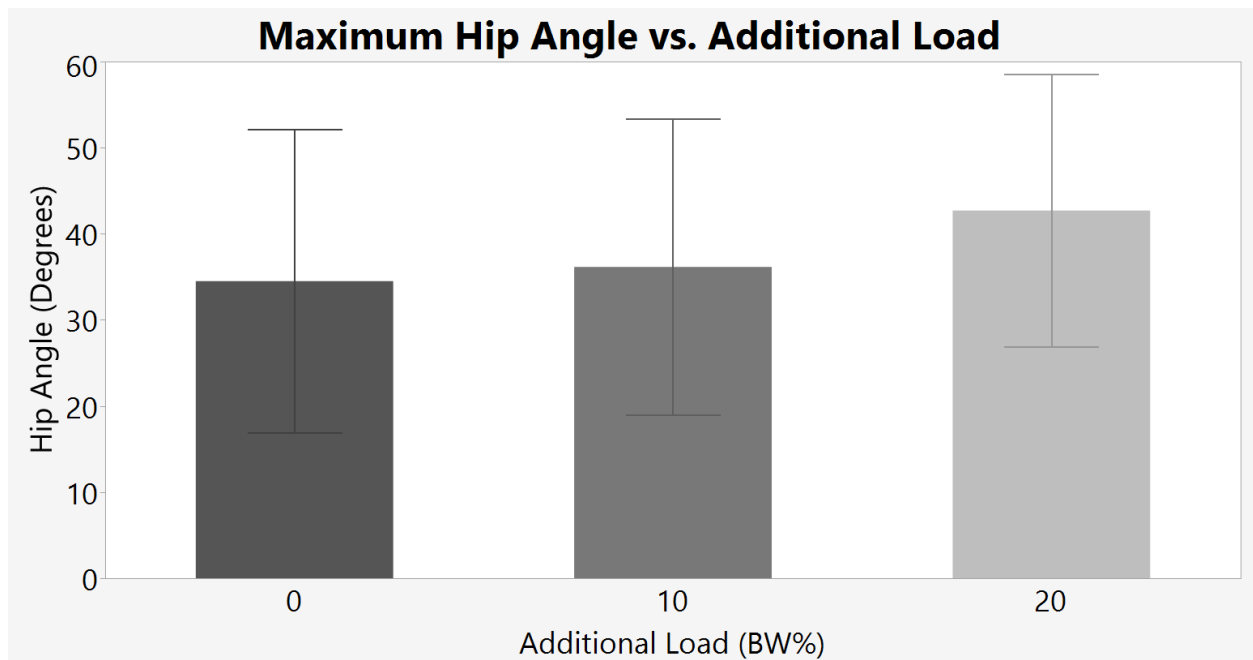


Figure 8: Mean minimum hip angle increased with greater backpack load. This data did not prove to be statistically significant.

Discussion

The goal of this study was to determine how progressively increasing backpack load influences sagittal-plane gait mechanics in healthy young adults, with particular focus on identifying a load threshold at which movement patterns begin to deviate from the baseline. Across all three aims, the results demonstrated consistent changes in trunk forward lean, knee angle, and hip angle load increased from 0% to 20% body weight.

The results of this study demonstrate numerical trends that generally align with existing biomechanical literature, though they lack the statistical significance reported in several previous works. Specifically, the observed numerical increase in mean trunk forward lean, from 7.96° at baseline to 12.07° at 20% BW, is consistent with the well-documented compensatory strategy used to maintain the body's center of mass over the base of support when carrying a posterior load. [1] [4] While research in children and military personnel has often identified clear thresholds for significant postural change at loads of 10-15% BW, this study did not statistically confirm a specific threshold for healthy young adults. [4] Regarding lower extremity kinematics, the numerical trend toward increased hip flexion under heavier loads mirrors significant findings reported in some studies. However, the overall lack of significant sagittal-plane changes in hip and knee angles in this study is in agreement with findings by Oberhofer et al. [6], who also reported no significant sagittal increases in hip or pelvis range of motion in a young adult cohort. These discrepancies likely reflect that healthy young adults possess greater musculoskeletal robustness and employ more diverse compensatory strategies than children, who typically exhibit more uniform movement adaptations. [8] While the small sample size and 2D video analysis are recognized limitations that reduced statistical power, the use of an ecologically valid outdoor walking protocol may provide a more realistic representation of everyday gait variability compared to highly controlled treadmill experiments, which often mask natural self-selected gait adjustments. Therefore, these findings highlight the inherent variability in young adult gait adaptation and suggest that universal loading thresholds established for other populations may not directly apply to healthy young adults.

This study incorporated several strengths that improved the clarity and reliability of the findings. First, the use of a controlled, repeated-measures design ensured that each participant served as their own comparison across the loading conditions, allowing subtle changes in kinematics to be detected. Second, the walking protocol allowed participants to move naturally, representing their everyday gait, capturing movement patterns that are more valid than treadmill-based assessments. This is relevant for load-carriage research, where self-selected adjustments in posture and step characteristics are part of the natural response to added loading. Third, the study examined multiple sagittal-plane variables simultaneously, including trunk lean, hip angle, and knee angle, allowing a more complete understanding of how the body redistributes motion when external load is added. Finally, the dataset included participants with a range of anthropometric characteristics, which contributed to a realistic representation of variability in human gait responses to load.

Despite these strengths, several limitations should be considered when interpreting the results. The most notable limitation is the small sample size, which reduced statistical power and

likely contributed to the absence of significant differences across loading conditions. Although trends were consistent with established biomechanical responses to load carriage, the variability within the sample limited the ability to detect group-level effects. Additionally, the use of 2D video analysis constrained the precision of joint angle measurements, particularly for the hip and knee, where out-of-plane motion and marker placement variability can influence angle estimates. A 3D motion capture system would have provided more detailed kinematic and kinetic information, including joint moments and ground reaction forces, which were beyond the scope of the study. The load increments (0%, 10%, 20% BW) were also relatively modest; larger or more finely spaced load levels may have produced clearer threshold relationships. Finally, participants walked at a self-selected speed, which enhances validity but introduces variability that may cover more subtle kinematic changes.

Healthy young adults demonstrate consistent gait adaptations as a backpack load increases, primarily through greater trunk forward lean, increased hip flexion, and altered knee positioning, even when those changes do not reach statistical significance at a group level. This study begins to demonstrate the importance of ongoing research with a larger test group size to further prove that increasing load does cause statistically significant results in the trunk lean data. Clinically, this study indicates that external loads approaching 20% body weight are sufficient enough to alter normal walking mechanics and increase mechanical demand on the musculoskeletal system. These findings suggest that clinicians should not rely solely on generalized “safe” recommendations but instead evaluate individual movement responses and postural compensation patterns when assessing load-related pain, overuse symptoms, or gait dysfunction. Patients who regularly carry occupational, military, or heavy academic loads may develop cumulative stress on the lumbar spine and lower joints even in the absence of obvious pathology. Early identification of excess forward trunk lean or abnormal lower-extremity mechanics during loaded gait may help guide preventative intervention, including load reduction and posture correction. Ultimately, this study supports the clinical importance of monitoring gait mechanics during load carriage rather than assuming healthy young adults uniformly tolerate moderate external loads without biomechanical consequence.

Acknowledgements

Artificial Intelligence was used as a tool to facilitate the research, discovering sources related to our topic. It was also prompted to help the team understand the context of these sources and how they can be used to benefit the study. Artificial Intelligence peer reviewed multiple paragraphs in the paper to correct grammar and sentence structure.

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